

SRILM n-gram LM In-Class Activity - Worked Solution (Study Guide)

Practice material for exam preparation; code in roll-colab.ipynb. Brown corpus, bigram unless stated; test set = brown-test.txt (305,056 words).

A. Toy corpus (5 sentences, 22 predicted tokens)

Q1 bigram: $P(S1 = \langle s \rangle \text{ Sam I do I like } \langle /s \rangle) = 3/5 \cdot 3/5 \cdot 1/5 \cdot 1/2 \cdot 2/5 \cdot 2/3 = 0.0096$; $P(S2 = \langle s \rangle \text{ Sam I am } \langle /s \rangle) = 3/5 \cdot 3/5 \cdot 2/5 \cdot 1/2 = 0.072$ -> **S2** is better. **Q2 unigram:** $S1 = 3.3e-5$, $S2 = 1.07e-3$ -> **S2**. **Q3:** 'linguistics' is unseen: $P(\text{linguistics}|\text{like}) = 0/3$, so unsmoothed unigram and bigram both give 0; add-one gives $3.6e-6$ (unigram) and $3.3e-5$ (bigram), using $V = 8$ word types. Per-word bigram terms: Sam|<s> .6, I|Sam .6, do|I .2, like|do .5, linguistics|like 0.

B. SRILM pipeline (verified): counts -> LM -> perplexity

```
ngram-count -text brown-train.txt -order 2 -write train.count -unk -lm train.lm -addsmooth 0 then ngram -lm train.lm -ppl brown-test.txt [-debug 2].
```

Cross-check: count of 'the' = 38,810 from both `grep -c '^the$'` on the token stream and the `.count` file.

C. Results (test perplexity)

Model	Smoothing	PPL	Zero-probs
Bigram	add-0 (+unk)	6270.9	39,856
Bigram	add-1	2493.2	0
Bigram	add-0.1	1208.8	0
Bigram	Good-Turing (default)	642.3	0
Bigram	Kneser-Ney (interp.)	617.9	0
Bigram	Witten-Bell (interp.)	416.2	0
Bigram	Modified KN (interp.)	362.2	0
Trigram	add-0 / Good-Turing / KN	6145.4 / 629.2 / 587.1	39,856 / 0 / 0
Bi, lowercase+no punct (KN)	own test text	916.4	0
Bi, no stopwords (KN)	own test text	1922.0	0
Uni+Bi mix, add .001, $\lambda=.7$	dev 581.0 (tuned)	681.3 (test)	0

D. Answers to Section 6

1. -inf: with `-addsmooth 0` (add-0 = plain relative frequency) `<unk>` has count 0, so every OOV test word (18,639 here) has $P=0$, $\log P=-\infty$; `-unk` only keeps `<unk>` in the vocabulary so OOVs map to it. Perplexity (6270.9) is misleading because zero-probability events are excluded. Fix: any real smoothing/discounting gives `<unk>` mass.

2. Smoothing: discounting methods (GT, KN, WB) beat add-delta by a wide margin (362-642 vs 1209-2493) and remove all zero probabilities; add-delta over-allocates mass to the huge unseen event space. Modified KN and Witten-Bell were best here; the ranking of KN vs WB may differ on other corpora/settings.

3. Trigram: KN 587 vs 618 for bigram (modest gain); unsmoothed trigram still has `-inf`. Longer context helps but sparsity grows (many unseen trigrams, so back-off/interpolation is essential).

4. Pre-processing: removing punctuation/stopwords changes the token stream itself, so perplexities across rows are *not comparable*: the remaining content words are harder to predict, which raises per-word ppl (916-4246), while total log-prob improves because there are fewer words. Compare only models evaluated on identical text.

5. Interpolation: trained unigram and bigram (add-delta) models, scored with `ngram -lm bi.lm -mix-lm uni.lm -lambda L` (L weights the main bigram LM). Grid over `add-delta` $\in \{.001, .01, .1, .5\}$ and `L` $\in \{.1, \dots, 1\}$; best on dev: `add=0.001`, `L=0.7`, dev ppl 581.0, test ppl 681.3. Note it is worse than KN (618) on test: simple add-delta with interpolation cannot match proper discounting.